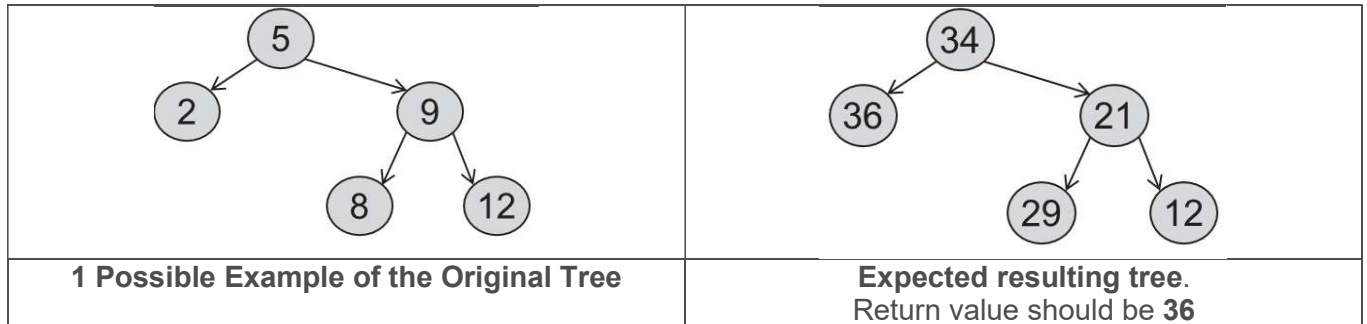


3. (12 pts.) **Binary Search Trees:** Assume you are given a valid binary search tree (may not be balanced), where each node is an instance of the Node struct given below. Write a **recursive** function (or recursive helper function) that will:
- return the sum of all the nodes and
 - update the value in each node with the **sum** of the all the values in the original tree greater than and including its own value. In the example below, 2's value becomes the sum of all values in the tree, 5's value becomes the sum of itself and all the values greater than it (i.e. 8, 19, 12), etc.



```
// assume appropriate #includes
struct Node {
    int val;   Node* left;  Node* right;
};

// prototype helper functions here, if needed
int sumLargeToSmallHelp(Node* root, int& csum);

int sumLargeToSmall(Node* root)
{
    int curr_sum = 0;
    sumLargeToSmallHelp(root, curr_sum);
    return curr_sum;
}
int sumLargeToSmallHelp(Node* root, int& csum)
{
    if(root == NULL) return 0;
    else {
        sumLargeToSmallHelp(root->right, csum);
        root->val += csum;
        csum = root->val;
        sumLargeToSmallHelp(root->left, csum);
    }
}
```